

Lorenzo Fant

lfant@ogs.it – +39 346 9521848 – Website: lorenzofant.github.io
Address: Via Gambini 47, 34141, Trieste (IT)

EXPERIENCES	<i>Researcher</i> , National Institute for Oceanography and Applied Geophysics - OGS, Trieste Topics: Macroecology, Data analysis and modeling	2024-now
	<i>PONTE Fellow</i> , Instituto Gulbenian de Ciencia, Lisbon Topics: Community metabolic scaling, Microbial functional ecology	2023-2024
	<i>Post-Doc</i> , Universidad de Granada Advisor: Miguel Angel Muñoz Topics: Theoretical ecology, Statistical physics, Stochastic processes	2022
	<i>Full-time employee</i> , Shoreline, Trieste Topics: Monitoring, conservation and restoration of marine ecosystems	2022
EDUCATION	<i>Ph.D in Physics</i> at SISSA & ICTP, Trieste Advisor: Jacopo Grilli Thesis: Stochastic ecology and evolution in bacterial communities Honors: cum laude	2017 - 2022
	<i>M.S. in Theoretical Physics</i> at Università di Trieste Final Grade: 110/110 cum laude Thesis Advisor: Fabio Finocchi - INSP (Paris) Erasmus at the ENS (Paris)	2014 - 2016
	<i>B.S. in Physics</i> , Università degli Studi di Trento Final Grade: 110/110 Thesis Advisor: Pietro Faccioli	2011 - 2014
EVENTS ORGANIZED	<i>IGC Symposium</i> , Biology across scales, Lisbon	2024
	<i>UGR Workshop</i> , Ecological Perturbations across scales, Granada	2023
GRANTS, FELLOWSHIPS AND AWARDS	<i>Young Researcher Prize - Biodiversity section</i> Gruppo 2003 per la Ricerca Scientifica	2025
	<i>Symposium organization grant</i> , IGC, Lisbon	2023
	<i>PONTE Fellowship</i> , IGC, Lisbon	2022-2024
	<i>Post-Graduate Fellowship</i> , SISSA, Trieste	2017
	<i>Full Scholarship</i> , Collegio L.Fonda, Trieste	2014-2016
	<i>Merit Prize</i> , Università degli studi di Trento	2016
PUBLICATION LIST	<i>PAF, Physics Deepening Course</i> , Università degli Studi di Trento	2011-2014
	<i>Ecological predictability emerges at the population level in phytoplankton communities</i> L. Fant, M. Klaassen, O. Mazzarisi, G. Ghedini bioRxiv: 2026.04.08.717202	2026

<i>Universal sublinear population growth density dependence unrelated to resource limitation</i>	2025
O. Mazzarisi , R. Droghetti, M. Carmichael, L. Ciandrini, M. C. Lagomarsino, M. Dal Bello, L. Fant, G. Ghedini, J. Grilli, G. Yvon-Durocher, D.C. Reuman bioRxiv: 2025.12. 30.697033	
<i>Fluctuating Environments Favor Extreme Dormancy Strategies and Penalize Intermediate Ones</i>	2025
J. Hidalgo , L. Fant , R. Rubio de Casas , M.A. Muñoz arXiv:2512.05856	
<i>Label invariance: a guiding principle for ecological models</i>	2025
T. Gibbs , O. Mazzarisi , L. Fant , R. An , J. Grilli , G. Barabs, C. Song BioRxiv: 10.1101/2025.07.15.664968	
<i>Biomass competition unifies individual and community scaling patterns</i>	2024
L. Fant, G. Ghedini Nature Communications 15	
<i>Stable Cooperation Emerges in Stochastic multiplicative growth</i>	2023
L. Fant, O. Mazzarisi, E. Panizon, J. Grilli Phys. Rev. E 108	
<i>Eco-evolutionary dynamics lead to functionally robust and redundant communities</i>	2021
L. Fant, I. Macocco, J. Grilli BioRxiv:10.1101/2021.04.02.438173	
<i>Variational Scheme to Compute Protein Reaction Pathways Using Atomistic Force Fields with Explicit Solvent</i>	2015
S. a Beccara, L. Fant, and P. Faccioli Phys. Rev. Lett. 114	
<i>Entangling macroscopic diamonds at room temperature: bounds on the CSL parameters</i>	2016
L. Fant et al. Phys. Rev. A 94	

TALKS

<i>Speaker @ AtlantEco Final Conference, Ponta Delgada</i>	2025
Title: Trophic interactions justify biodiversity's variation	
<i>Speaker @ Scientific Council evaluation, Trieste</i>	2025
Title: Trophic interactions justify biodiversity's variation	
<i>Speaker @ Complexity of Life: bridging scales and biological systems, Graz</i>	2024
Title: Biomass competition unifies individual and community scaling patterns	
<i>Speaker @ Ecological Perturbations across scales, Granada</i>	2023
Title: Niche dispersal: an evolutionary phase transition	
<i>Invited Speaker @ Ergodicity Economics, Lisbon</i>	2023
Title: The ergodicity economics perspective on cooperation	
<i>Speaker @ Ergodicity Economics, Online</i>	2022

	Title: Cooperation in Stochastic multiplicative-growth environments is evolutionary stable	
	<i>Invited Seminar @ Max Plank Institute for Mathematics, Leipzig</i>	2021
	Title: Cooperation in Stochastic multiplicative-growth environments is evolutionary stable	
	<i>Speaker @ SMEEB, Venezia</i>	2021
	Title: Eco-evolutionary dynamics lead to functionally robust and redundant communities	
	<i>Poster @ QLIFE - Quantitative Ecological Genomics in the Tara Ocean</i>	2021
	Title: Eco-evolutionary dynamics lead to functionally robust and redundant communities	
JOURNAL REFEREE	<i>Physical Review Letters, Physical Review X Life, Physical Review Research, Physical Review E, Marine Ecology Progress Series</i>	
COMPUTER SKILLS	Languages: <i>Python, Julia</i> <i>SQL, C, Html</i>	Advanced Intermediate
LANGUAGE SKILLS	<i>Italian</i> <i>English</i> <i>Spanish, French</i> <i>Slovenian</i>	Mothertongue IELTS 7.5 Intermediate Basic
EXTRA CURRICULAR ACTIVITIES	<i>Students Representative</i> Academic Senate, SISSA Department of Physics, Trento Liceo G.B. Benedetti, Venezia <i>WWF marine guide</i> , AMP Miramare, Trieste <i>Consultant Physicist</i> , ERREQUATTRO s.r.l.	2018 - 2020 2012 - 2014 2010 2018 - 2022 2015
LICENSES	<i>Boating</i> , without limitations <i>Scuba Diving</i> , PADI Advanced	
REFERENCES	<i>Giulia Ghedini</i> , IGC, Lisbon, Portugal – gghedini@igc.gulbenkian.pt <i>Jacopo Grilli</i> , ICTP, Trieste, Italy – jgrilli@ictp.it <i>Marco Cosentino Lagomarsino</i> , IFOM, Milan, Italy – marco.cosentino-lagomarsino@ifom.eu <i>Miguel Angel Muñoz</i> , UGR, Granada, Spain – mmunoz@onsager.ugr.es	